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**Faculty of Engineering & Technology Electrical &
Computer
Engineering Department
Analog Electronics ENEE 2360
Report of Course Project
DC Power Supply & Automatic Light**

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Abstract:

In this project in part one we are learning how to Design, simulate and implement an ac/dc rectifier with center tapped transformer . As for part two, which aims to employ electronic devices to regulate the lamp with the level of light that automatically turns on at dark and off throughout the day, is an automatic light control device. The following methodology will be used to complete this project utilizing the electrical components listed: transistor, resistors, capacitor, diode, and transformer. First, analyze the automatic lighting control circuit. Next, design and simulation the circuit using PSpice software.

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1. Theory

1.1 Center tapped Full Wave rectifier:

The Center Tapped Full Wave Rectifier employs a transformer with the secondary winding AB tapped at the centre point C. It converts the AC input voltage into DC voltage. The two diode D_1 , and D_2 are connected in the circuit as shown in the circuit diagram below.

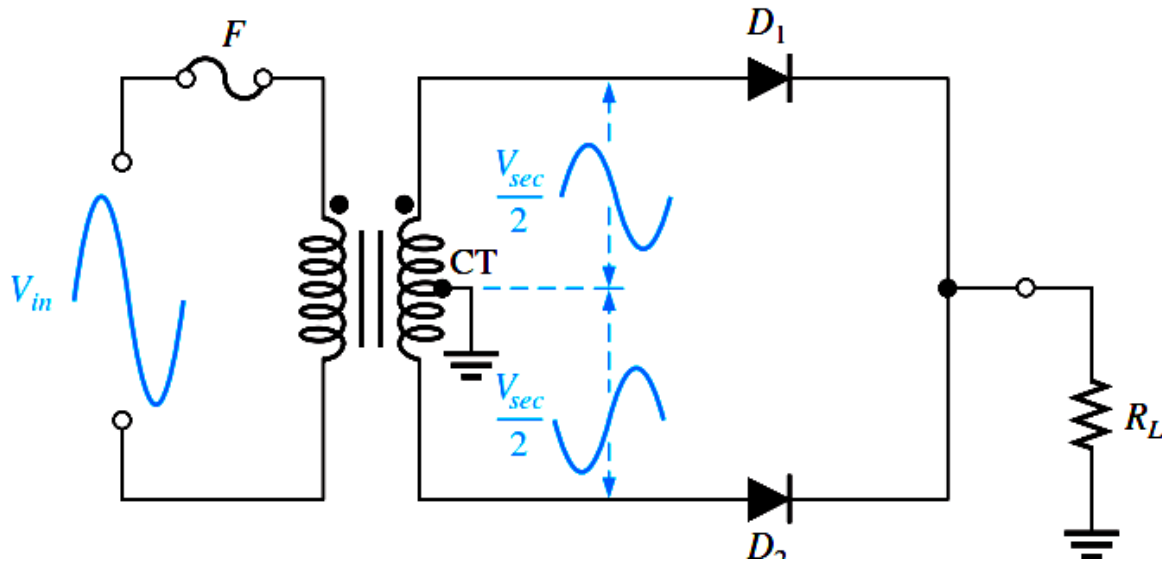


Figure 1: Center tapped Full Wave rectifier

1.1.1 Operation of the Center Tapped Full Wave Rectifier

When AC supply is switched ON the alternating voltage, V_{in} appears across the terminals AB of the secondary winding of the transformer. During the positive half cycle of the secondary voltage, end A becomes positive, and end B becomes negative. Thus, the diode D_1 becomes **forward biased**, and diode D_2 becomes **reverse biased**.

The two diodes conduct simultaneously. Therefore, when the diode D_1 conducts, the diode D_2 does not conduct and vice versa.

When the Diode D_1 is conducting, the current (i) flows through the diode D_1 load resistor R_L (from M to L) and the upper half of the secondary winding as shown in the circuit diagram marked by the red color arrowheads. During the negative half-cycle, the end B becomes positive, and end A becomes negative. This makes the diode D_2 forward biased, and diode D_1 reverse biased.

When the diode D_2 conducts while the diode D_1 does not. The current (i) flows through the diode D_2 load resistor R_L (from M to L) and the lower half of the secondary winding as shown by the red dotted arrows.



The current flowing through the load resistor R_L is in the same direction (i.e., from M to L) during both the positive as well as the negative half cycle of the input. Hence, the DC output voltage ($V_{out} = i R_L$) is obtained across the load resistor.

The wave diagram of the input voltage, the current flowing through the load, and the output voltage developed across the load is shown in the figure below:

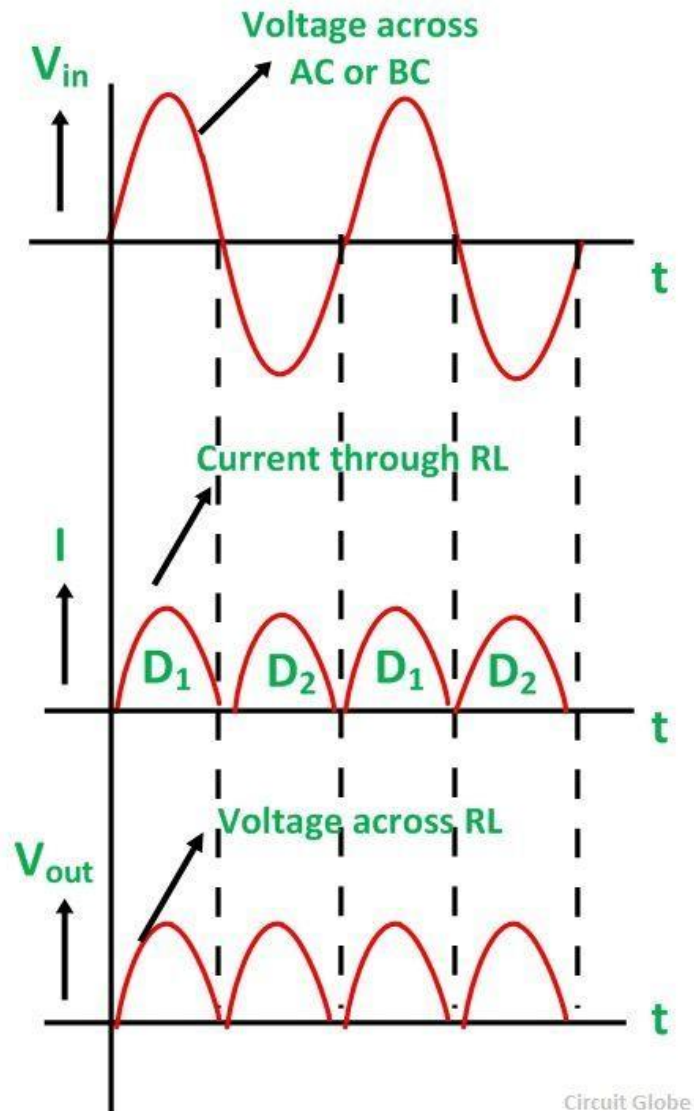


Figure 2 : The wave diagram



1.2 Automatic Light /Lamp controller

It is considered an automatic lighting control unit where the lighting turns on automatically when the lighting turns on automatically. This system uses the transistor as an **on/off** switch the system makes use of the transistor work in the local area and cut off to automatically turn on and off the lights. This system works as a sensor, as it does not require manual control, the system detects light itself. When the light is light or light, the light value will show light and show light automatically. Measure it on a (**variable resistor**) to control how dark the light should be.

1.2.1 Features:

- It is a dark detector circuit based on LDR and a transistor which automatically switches ON and OFF the street light system.
- It automatically switches ON street lights when the sunlight goes below the visible region of our eyes. (e.g. in the evening after sunset).
- It automatically switches OFF the lights when sunlight falls on it (i.e. on LDR) e.g. in the morning, the sensor called LDR (Light Dependent Resistor) senses the light just like our eyes and deactivates the circuit.

1.2.2 Advantages:

- The automatic operation of street light controlling systems help to reduce the energy consumption as compared to the manually operated street light controlling operations. This is because there is a delay in the earlier switching operations both in morning (during sunrise) and evening (during sunset).
- On sunny and rainy days, ON and OFF time is noticeably differ which is one of the major disadvantages of using timer circuits or manual operation for switching the street light system.

1.2.3 Components Required:

- LDR – Light Dependent Resistor
- Transistor.
- Resistor
- Light emitting diode (LED)
- Connecting wires
- Power supply

✚ 2. Procedure & Results

2.1 Part 1 (DC Power Supply):

2.1.1 Design in PSpice :

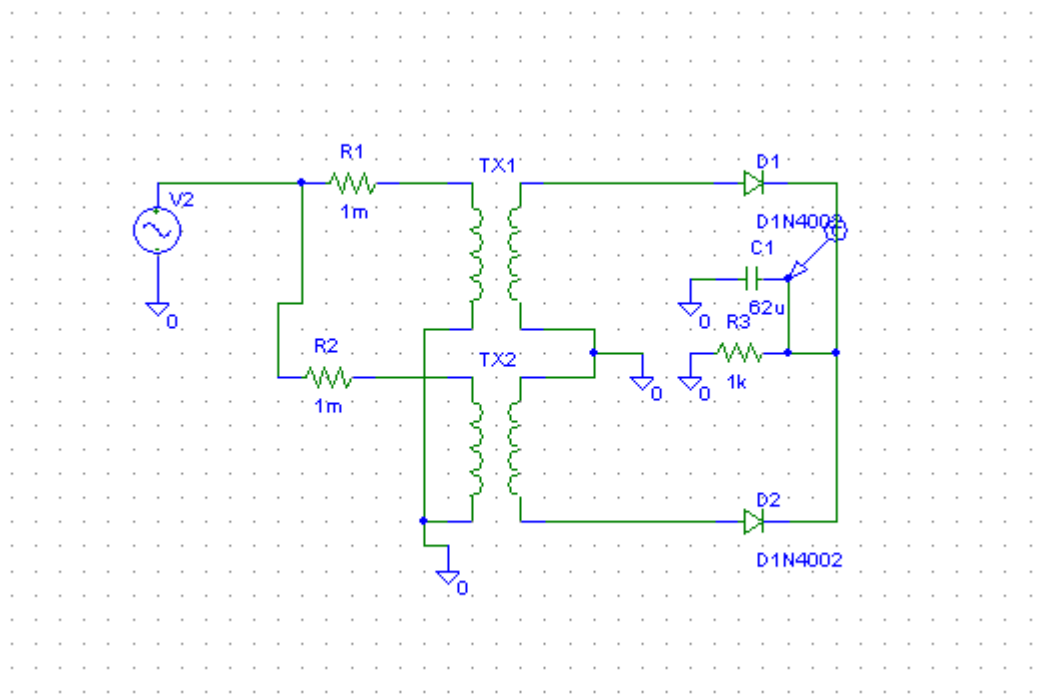


Figure 3 : our circuit in PSpice



2.1.2 calculation

Calculations:-

$$\textcircled{1} \quad V_{i,p-p} = \sqrt{2} V_{rms}$$
$$= \sqrt{2} \times 220$$

$$V_{i,p-p} = 311.1 \text{ volt}$$

$\textcircled{7}$ To Find $C \Rightarrow$

$$\textcircled{2} \quad r = \frac{V_{rL,rms}}{V_{o,dc}}$$

$$0.05 = \frac{V_{rL,rms}}{9}$$

$$C = \frac{V_{om}}{2\pi R V_{rL,p-p}}$$

$$C = 6.2 \times 10^{-5} \text{ F}$$

$$V_{rL,rms} = 0.45 \text{ volt}$$

$$\textcircled{3} \quad V_{rL,p-p} = 2\sqrt{3} V_{rL,rms}$$
$$= 2\sqrt{3} \times 0.45$$

$$V_{rL,p-p} = 1.59 \text{ volt}$$

$$\textcircled{4} \quad V_{om} = V_{o,avg} + \frac{1}{2} V_{rL,p-p}$$

$$V_{om} = 9.78 \text{ volt}$$

$$\textcircled{5} \quad V_{ms} = V_{om} + 0.7$$

$$V_{ms} = 10.48 \text{ volt}$$

$\textcircled{6}$ To find L_1 and L_2 :-

$$\left(\frac{V_{mp}}{V_{ms}} \right)^2 = \frac{L_1}{L_2} \Rightarrow \text{assume}$$

$$L_2 = 1 \text{ mH}$$

$$L_1 = 88.1 \text{ mH}$$

Figure 4 : calculation part 1



2.1.3 Simulation

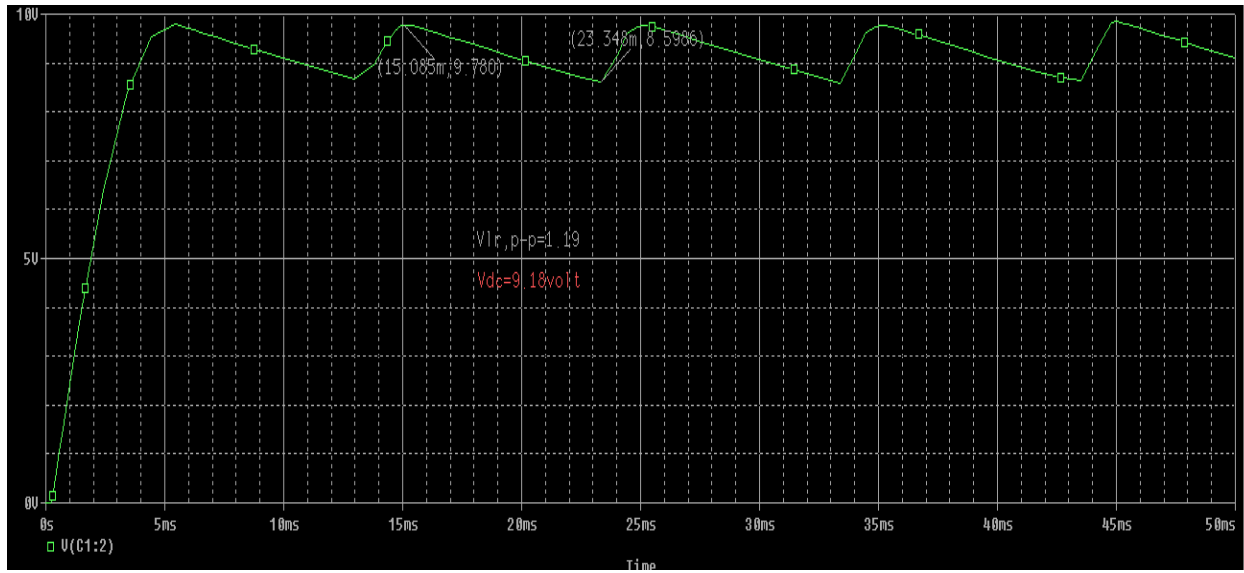


Figure 5 :Simulation part 1

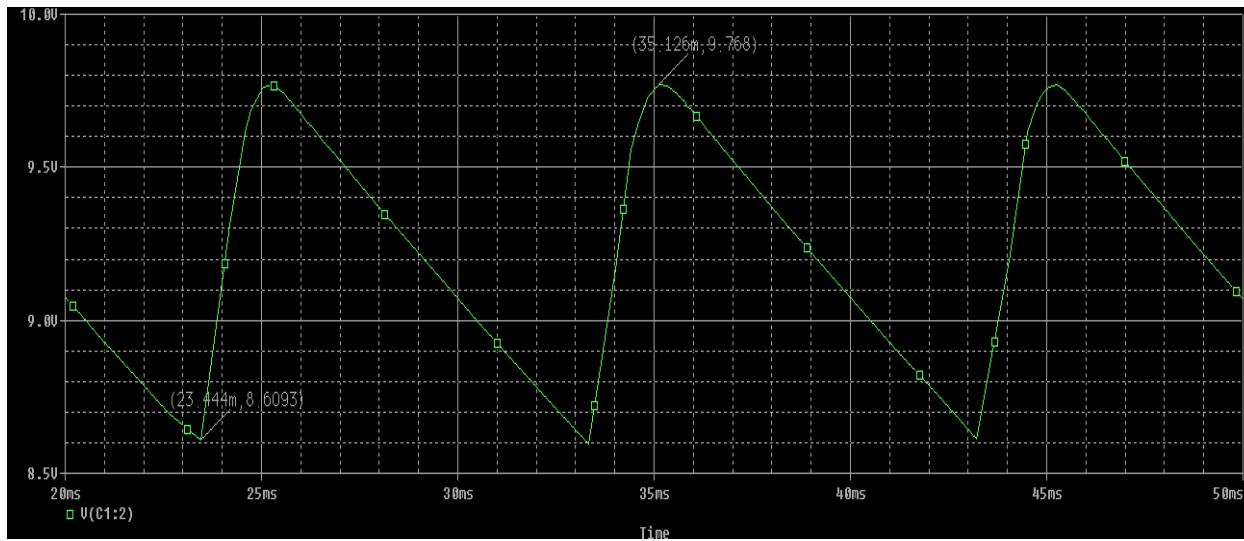


Figure 6: zoom



2.1.4 Practical Results

When networking the circuit with the assistant, we used the values of the transformer attached to it (12v) and used the value of the capacitor 94 ; Which led to these values and results:

$$V_{rms} = 12 * \sqrt{2} = 17$$

$$V_{dc} = \frac{2*17}{3.14} = 10.8$$

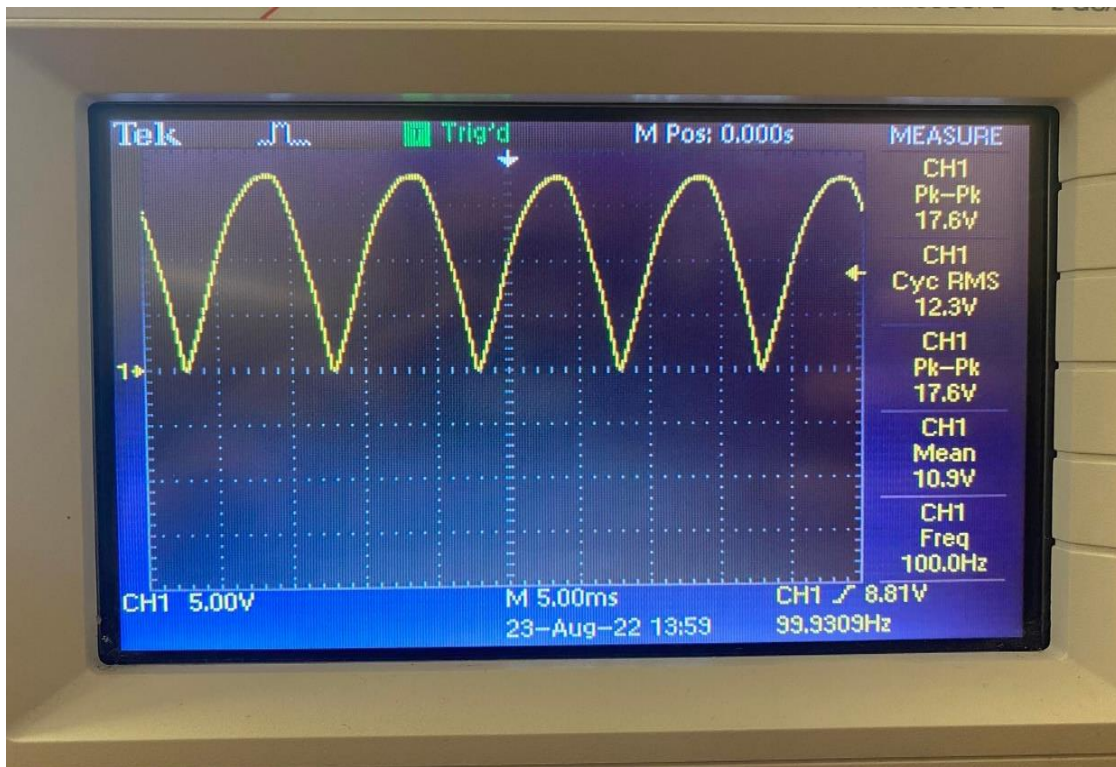


Figure 7: practical before cap

Before capacitor :

$$V_{dc} = 10.8$$

$$V_{ac} = 5.47$$

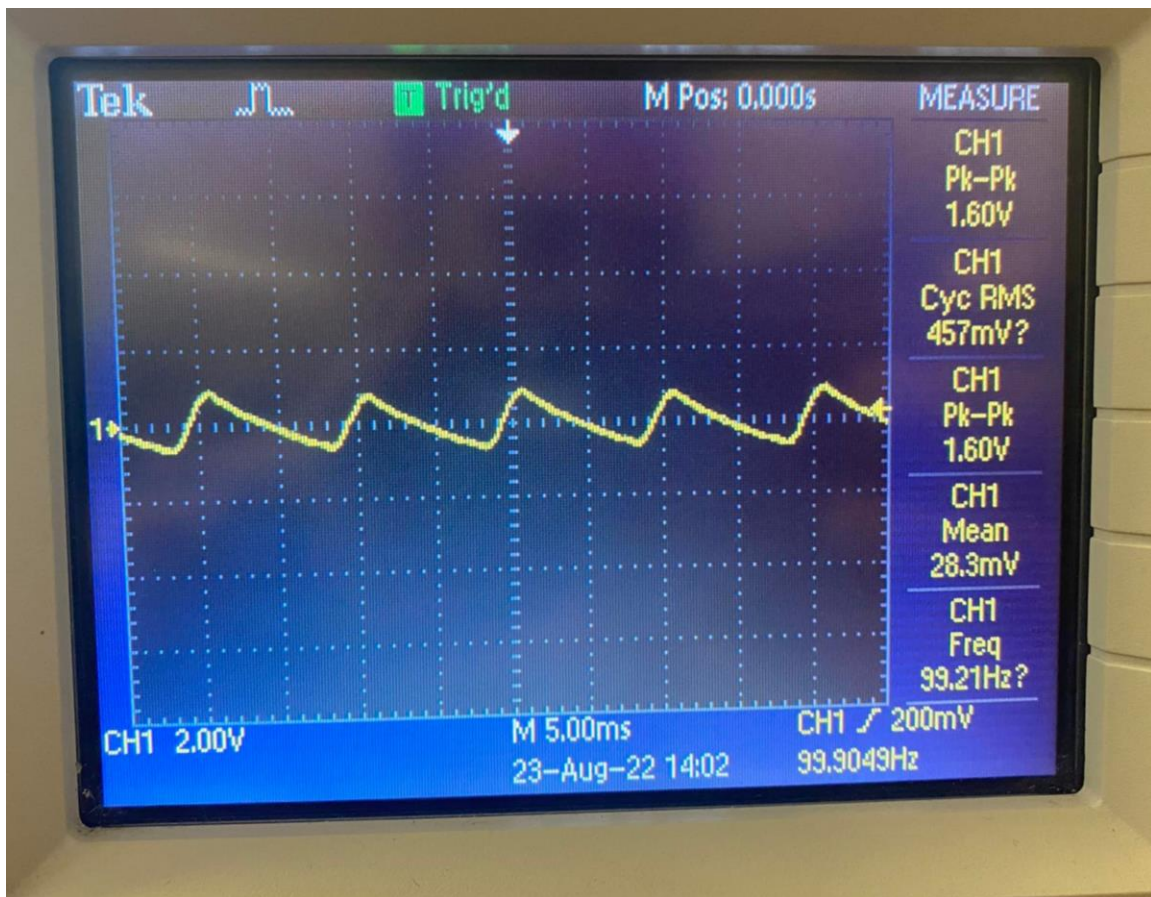


Figure 8:practical after cap

After capacitor:

$V_{dc} = 16.38$

$V_{ac} \text{ (p-p ripple)} = 1.6$



2.2 Part two (Automatic Light)

2.2.1 Design in PCpice :

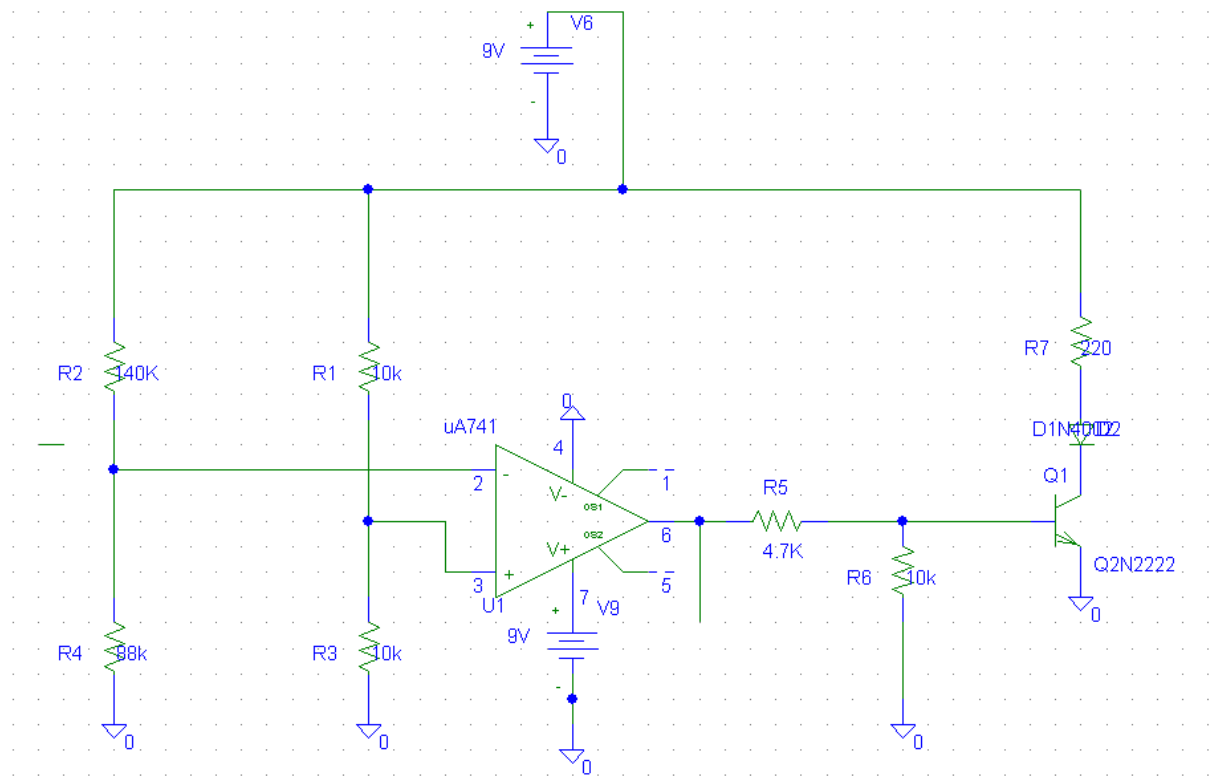


Figure 9: Circuit part 2

In this part we set the LDR to 3K (Light) so that the diode is off and the current is approximately 0. Then, we changed it to 140K (Dark) so that the diode is on and there is current passes through the diode.

(The values are shown in calculation part)



2.2.2 calculation

Calculations

- LDR Low (Light)

$$V_o = -V_{sat}$$

$$V_- > V_+$$

$$\frac{R}{R+3} \cdot 9 > \frac{10k}{10k+10k} \cdot 9$$

$$\frac{R}{R+3} > 0.5$$

$$\boxed{R > 3}$$

- LDR High (Dark)

$$V_o = +V_{sat}$$

$$V_- < V_+$$

$$\frac{R}{R+140} \cdot 9 < 4.5$$

$$\boxed{R < 140}$$

The range of R is :

$$\boxed{3 < R < 140}$$

Figure 10 : Calculation part 2



2.2.3 Simulation

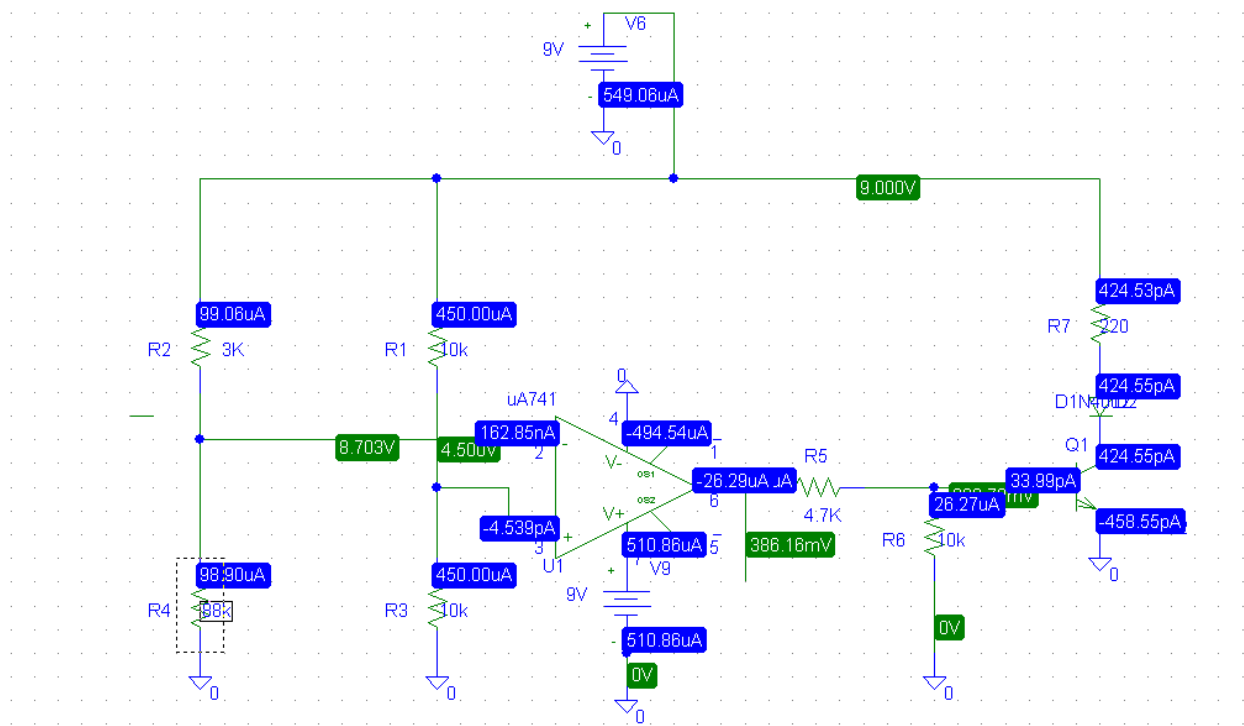


Figure 11: Running when $R_{LDR} = 3k$

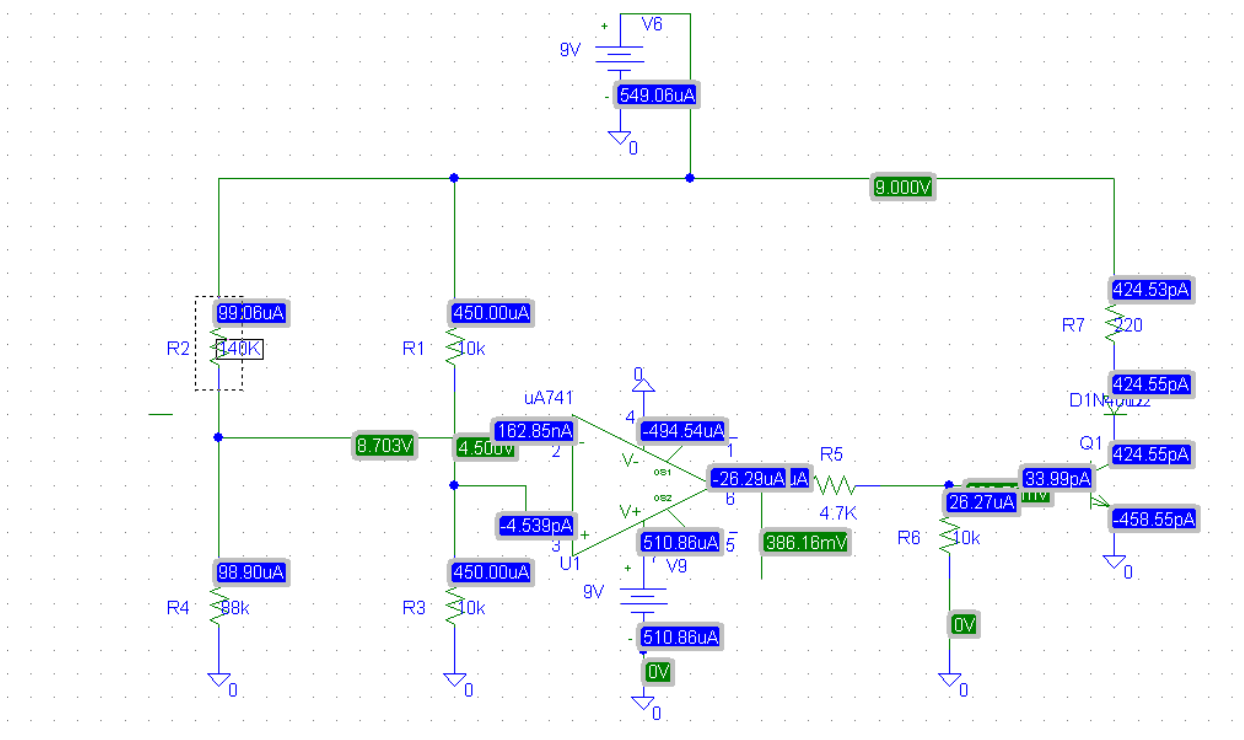


Figure 12: Running when $R_{LDR} = 140k$

3. Conclusion

In this project, we worked with electronic components like transistors , diodes, capacitors and Op-amp and we learnt how to link them in a circuit using PSpice tools and simulate it to acquire the values we need. In part one we create a DC power source, by linked diodes, capacitors, and transformers to get a specified Vdc (9v) and we got approximate value (9.18v) from the graph to construct a full wave center tapped rectifier. Then in part two, we made contact with the transistors that served as switches to form Automatic Light . By using range of Rldr (3k-140k) to find the variable resistance. In genral, the goals of the project were successfully met, and it was highly beneficial.



5. References

[1] <https://circuitglobe.com/center-tapped-full-wave-rectifier.html>

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[2] <https://www.electricaltechnology.org/2013/04/automatic-street-light-control.html>

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